

No. of Functional Features Included in the Solution

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Load and manage the flood prediction dataset	Data Collection	Done	High
2	Data Visualization	Perform univariate, multivariate and descriptive analysis	Data Analysis	Done	High
3	Data Preprocessing	Handle missing values, outliers, categorical values and feature scaling	Data Preprocessing	Done	High
4	Machine Learning Models	Train Decision Tree, Random Forest, KNN and XGBoost models	Model Building	Done	High
5	Model Comparison	Compare all trained models and select the best-performing model	Model Evaluation	Done	High
6	Flood Prediction	Generate flood predictions using the trained model	Prediction Engine	Done	High
7	Flask Web Application	Provide a user-friendly interface for prediction	Application Building	Done	High
8	Model Deployment	Deploy the application for public access	Deployment	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	7
Additional / Nice-to-Have Features	1
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	6	Home Page, Parameter Input Form, Prediction Result Pages (Chance / No Chance), Analytics Dashboard, History Logs.
2	Backend / Logic	3	Input parsing, real-time feature scaling, model classification/probability evaluation and offline model training.
3	Database / Storage	2	SQL user tables, SQL history records, and serialized model/scaler binaries.
4	API / Integration	1	Programmatic cloud deployment management.
5	Security / Authentication	3	Password cryptographic hashing, private routing controls and form float value error handling.